Experiment 2

Implementation of Double Data Encryption Standard (2DES)

Aim

To implement the Double Data Encryption Standard (2DES) algorithm for encrypting and decrypting plaintext using two different secret keys.

Description

Double Data Encryption Standard (2DES) is an extension of the DES algorithm that applies the encryption process twice using two different keys. The plaintext is first encrypted with the first key, and the resulting ciphertext is encrypted again with the second key. During decryption, the ciphertext is first decrypted with the second key and then with the first key to recover the original plaintext. Using two keys increases the security compared to a single DES operation.

Algorithm

1. Start the program.
2. Read the plaintext from the user.
3. Read two secret keys.
4. Encrypt the plaintext using the first key.
5. Encrypt the result again using the second key.
6. Display the encrypted text.
7. Decrypt the encrypted text using the second key.
8. Decrypt the result using the first key.
9. Display the original plaintext.
10. Stop the program.

Code

```
plaintext = input("Enter Plain Text: ").upper()

key1 = int(input("Enter Key 1: "))

key2 = int(input("Enter Key 2: "))

encrypted = ""

for ch in plaintext:
```

```
    if ch.isalpha():
        x = (ord(ch) - 65 + key1) % 26
        y = (x + key2) % 26
        encrypted += chr(y + 65)
    else:
        encrypted += ch
print("Encrypted Text:", encrypted)
decrypted = ""
for ch in encrypted:
    if ch.isalpha():
        x = (ord(ch) - 65 - key2) % 26
        y = (x - key1) % 26
        decrypted += chr(y + 65)
    else:
        decrypted += ch
print("Decrypted Text:", decrypted)
```

Output: